

Week 9 — Day 1 Labs

Lab 1: Product Model + Lab 2: Strengthen the Product Model

Lab 1 — Required Explanation

1 — Create the app and install it

Create the **catalog** app and add it to **INSTALLED_APPS**. This allows Django to recognize the app as part of the project.

2 — Define Product

Define the **Product** model inside **catalog/models.py**. The model represents the product entity from the exercise.

3 — Required name and unique SKU

Add the required **name** field and a unique **sku** field. The SKU must not be repeated between products.

4 — Optional description

Use **blank=True** for description so the description can be left empty in form validation.

5 — Category.TextChoices

Create **Category.TextChoices** with the three options shown in the lab: Electronics, Food, and Accessory.

6 — Price and non-negative stock

Use a **DecimalField** for price and a non-negative integer field for stock.

7 — Active and timestamps

Add **is_active**, **created_at**, and **updated_at** fields as required by the lab.

8 — __str__ and check

Add **__str__()**, then run **python manage.py check** to verify the project.

Lab 2 — Required Explanation

1 — `is_available()`

Returns True when the product is active and stock is greater than zero.

2 — `inventory_value()`

Calculates inventory value using $\text{price} \times \text{stock}$.

3 — Default ordering

Orders by category first, then name.

4 — Verbose names

Defines clear singular and plural names: Product and Products.

5 — Composite index

Adds an index covering category and `is_active` together.

6 — Price constraint

Requires $\text{price} \geq 0$ using a named `CheckConstraint`.

7 — Stock constraint

Requires $\text{stock} \geq 0$ using a named `CheckConstraint`.

8 — Run check and explain

Run `python manage.py check` and explain each design choice made in the model.

Exit Ticket — Lab 1

Why does price use `DecimalField` and description use `blank=True`?

Price uses `DecimalField` because product prices need decimal precision. Description uses `blank=True` because the lab defines description as optional and allows it to be left empty during form validation.

Exit Ticket — Lab 2

Why does non-negative stock need both?

Friendly validation gives users understandable feedback, while the database constraint protects stored data at the database layer.